PHD STUDENT · COMPUTER SCIENCE · UNIVERSITY OF SOUTH CAROLINA

Kasra Korminejad

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ABOUT

I do research at the intersection of AI and biology. I design and train AI models that solve biological problems, from finding the genes that drive a response in RNA-seq data to predicting how proteins fold.

Working across both fields lets me turn biological questions into machine learning problems, and turn model outputs back into biological insight. The long-term goal is research that helps us understand and treat diseases like cancer. Before my PhD I spent several years as a software engineer, so I build models that are efficient and ready to deploy, not just accurate on paper.

EDUCATION

Ph.D. in Computer Science — Jan 2025 – 2031 (expected)

University of South Carolina · AI Institute of South Carolina · Advisor: Dr. Homayoun Valafar. Research focus: AI and bioinformatics for genomic and proteomic data.

B.Sc. in Computer Engineering — 2016 – 2021

Ferdowsi University of Mashhad

SKILLS

Machine learning	Deep learning, CNNs, LSTMs, Transformers, Attention models, Reinforcement learning, Generative sequence models
Computational biology	Bioinformatics, Genomics, Molecular biology, Bulk RNA-seq, Differential expression (DESeq2), Gene networks, Protein structure prediction, AlphaFold2
Molecular dynamics	VMD, NAMD
Programming	Python, C, C++, Java, JavaScript, SQL
ML & data tools	PyTorch, Hugging Face, NumPy, pandas
Engineering	Linux, Docker, Git, AWS, Azure, FastAPI, Flask, CI/CD

RESEARCH

TRANSCRIPTOMICS ONGOING

Machine learning for high-dimensional RNA-seq data

I train machine learning models that find meaningful patterns in bulk RNA-seq gene expression data, alongside bioinformatics methods such as differential expression analysis. The current application is a rat study of how kynurenine changes gene expression.

> **The core challenge is p ≫ n.** Tens of thousands of genes are measured on a small number of samples, so models have to be regularized and interpretable to find real biological signal instead of noise.

SYSTEMS BIOLOGY ONGOING

Gene networks across species

Genes work in networks, not alone. This project models the relationships and interactions between genes, and how those relationships compare between different species.

STRUCTURAL BIOLOGY ONGOING

Protein structure prediction beyond AlphaFold

AlphaFold changed the field, but parts of protein structure prediction are still unsolved, including accurate side-chain placement and regions predicted with low confidence. As one of my main PhD projects, I'm extending AlphaFold2 with predicted inter-atomic bond angles as structural priors, aiming for structures reliable enough for ligand discovery and drug design.

PROJECTS

DEEP LEARNING · SLEEP-EDF

Near real-time sleep stage classification

A deep learning model that classifies sleep stages from EEG on the Sleep-EDF dataset. A CNN extracts features from each 30-second epoch, then an LSTM and multi-head attention layers learn patterns within and across epochs. It works with a 5 to 10 minute delay instead of needing a full night of recording, and it is small enough to deploy on edge devices.

79.4% macro F1 **84.7%** accuracy **614k** parameters

GENOMICS · ALGORITHMS

MRA: every tandem repeat in one scan of DNA

Tandem repeats make up about 3% of human DNA, mutate easily, and are linked to diseases such as neurodegeneration. Existing tools each cover one class of repeat. The Multi-head Reader Arm (MRA) algorithm finds exact and inexact repeats of every pattern length in a single linear-time pass: a base head and one head per pattern length move along the sequence together, comparing characters as they go.

On human chromosomes 1 to 5 (181 to 249 million bases), MRA ran faster than TRF, Mreps, Perf and Kmer-SSR while using almost no extra memory. Built at Ferdowsi University with Stanford University collaborators.

PUBLICATIONS

A Novel Multi-head Algorithm to Discover All Tandem Repeats in One Scan of DNA Sequences
Naghibzadeh, M., **Korminejad, K.**, Krishnan, V., Bahmani, A.
2023 · doi:10.21203/rs.3.rs-2879921/v1

EXPERIENCE

Jan 2025 – now
Graduate Research Assistant
University of South Carolina · AI Institute of South Carolina · Advisor: Dr. Homayoun Valafar
Machine learning for RNA-seq data, cross-species gene networks and protein structure prediction.

Jan 2024 – Sep 2024
Back-End Software Engineer
ParstechAI
- Built and maintained distributed back-end systems for large-scale chatbot infrastructure.
- Refactored core services around clean-architecture principles and helped design a more resilient architecture.

Nov 2020 – Dec 2023
Research Assistant, Distributed Systems Lab
Ferdowsi University of Mashhad
- Co-developed MRA, a one-scan algorithm for tandem repeat discovery, with collaborators at Stanford University.
- Benchmarked against state-of-the-art tools on whole human chromosomes.

Jun 2022 – Sep 2022
Software Engineer
Arta Exchange (crypto aggregator)
Real-time data and warehouse services for arbitrage analytics, with high-frequency pipelines synchronized to the millisecond across exchanges.

Jan 2021 – Jun 2022
Research Assistant, Machine Vision Lab
Ferdowsi University of Mashhad
Deep learning models for MRI-based hand strength estimation and injury classification.

Sep 2020 – Aug 2021
Junior Full-Stack Developer
Dezigno (e-commerce)
Front-end and back-end microservices for a print-on-demand platform: product design, orders and payments.

TEACHING & OUTREACH

Summer
AI & Machine Learning Bootcamp Instructor
High school students · Richland County, South Carolina
Taught AI and machine learning to high school students during a one-week summer bootcamp.